AI-Powered Agriculture Crop Advisory Assistant

Overview

Agricultural professionals regularly handle recurring situations involving crop selection, soil conditions, irrigation, fertilizer usage, weather variations, and crop productivity. Over time, agricultural organizations accumulate thousands of agricultural records containing valuable information about soil characteristics, weather conditions, crop types, farming inputs, production, and crop yields.

However, manually searching through historical agricultural records to identify similar farming conditions and successful crop practices can be time-consuming and inefficient.

Agricultural professionals often ask questions such as:

• A farmer has specific soil nutrient levels, weather conditions, and available farming inputs. Which crops have performed well under similar conditions?

• Multiple farms are growing the same crop under different soil and weather conditions. What factors could explain the difference in yield?

• A farmer wants to select an alternative crop for a particular season and region. Which crops have historically performed well under similar conditions?

Instead of manually searching through historical agricultural records, agricultural professionals should be able to quickly retrieve similar farming cases and crop recommendations.

Build a semantic retrieval system that allows agricultural professionals to enter natural-language queries and retrieve the most relevant historical farming cases along with crop, yield, and agricultural recommendations.

Requirement 1 (Basic)

• Basic RAG for agricultural case similarity search

• Hybrid search (keywords + semantic)

• Crop and agricultural condition classification agent

• Metadata filtering (crop, season, state, soil conditions, weather conditions)

• Basic crop recommendation

• Input validation guardrails

• Case prioritization

Expose the core functionality through API endpoints.

Requirement 2 (Advanced)

• DeepEval for recommendation quality evaluation

• Crop suitability risk prediction

• LLM-as-judge for crop recommendation validation

• Multi-agent agricultural advisory collaboration

• Crop yield prediction

• Soil, weather, and farming-input impact analysis

• Feedback loop for continuous improvement

Build a simple front-end interface.

Deliverables

1. Architecture Diagram

• Include a high-level view of your system in JPEG or PDF.

• Show how data is processed from the provided dataset through to the user's query response.

2. Design

• Articulate system design decisions taken and the trade-offs made.

3. Full Executable Code (Microservice)

• Focus on clarity and modularity.

• Provide a README explaining:

▪ Project setup (how to install or run)

▪ Sample usage (e.g., a test query and the resulting crop recommendations)

4. Panel Presentation (10 minutes)

▪ Demo of your working solution outlining the design and approach (8 minutes).

▪ Q&A (2 minutes).

Dataset Name

Crop Yield Data with Soil and Weather Dataset

Primary Link

Crop Yield Data with Soil and Weather Dataset

https://www.kaggle.com/datasets/anshumish/crop-yield-data-with-soil-and-weather-dataset

Alternative Link

FAOSTAT Crop Production, Yield and Harvested Area

https://www.fao.org/faostat/en/#data/QCL

Format

CSV

Key Fields

Crop, Crop_Year, Season, State, Area, Production, Yield, Fertilizer, Pesticide, N, P, K, pH, Temperature, Rainfall, Humidity

Reason

Historical crop, soil, weather, fertilizer, pesticide, production, and yield information useful for agricultural case retrieval, crop recommendation, crop suitability analysis, yield prediction, alternative crop recommendation, and agricultural decision support.